

NARPAN TABLET

DESCRIPTION:

NARPAN TABLET: A blue, oval tablet with marking 'DUO' and scored '861'

COMPOSITION:

Each tablet contains Naltrexone HCl 50 mg

PHARMACODYNAMICS: Naltrexone HCl is a pure opioid antagonist. It markedly attenuates or completely blocks reversibly, the subjective effects of IV administered opioids. (The term opioid is used to describe (1) classic morphine-like agonists and (2) analgesics possessing agonist and antagonist activity (e.g. buprenorphine, nalbuphine and pentazocine.) When co-administered with morphine, on a chronic basis, Naltrexone blocks the physical dependence to morphine and presumably other opioids. Naltrexone has few, intrinsic actions besides its opioid blocking properties. However, it does produce some pupillary constriction. The administration of Naltrexone is not associated with the development of tolerance or dependence. In subjects physically dependent on opioids, Naltrexone will precipitate withdrawal symptoms. 50mg of Naltrexone will block the pharmacologic effects of 25mg IV administered heroin for periods as long as 24 hours. Doubling the dose of Naltrexone provides blockade for 48 hrs and tripling the dose of Naltrexone provides blockade for about 72 hours. While mechanism of action is not fully understood, evidence suggests that Naltrexone block the effects of opioids by competitive binding (i.e. analogous to competitive inhibition of enzymes) at opioid receptors. This makes the blockade produced very extensive.

PHARMACOKINETICS: After oral administration, Naltrexone is subject to extensive "first pass" hepatic metabolism (its major route of elimination) with approximately 95% of the absorbed drug being converted to several metabolites. The major metabolite, 6- β -naltrexol like Naltrexone is believed to be a pure antagonist and may contribute to the pharmacological blockade of opioid receptors. A minor metabolite is 2-hydroxy-3-methoxy-6- β -naltrexol. Naltrexone and its metabolites are also conjugated to form additional metabolic products. Naltrexone and its metabolites are excreted primarily by the kidney, with fecal excretion being a minor elimination pathway. The urinary excretion of unchanged Naltrexone accounts for <1% of an oral dose; urinary excretion of unchanged and conjugated 6- β -naltrexol accounts for approximately 38% of an oral dose. The pharmacokinetic profile of Naltrexone suggests that Naltrexone and its metabolites undergo enterohepatic recycling. Following the administration of 50mg Naltrexone tablets to 24 healthy adult male, the C_{max} for Naltrexone and its major metabolite, 6- β -naltrexol were 8.6 and 99.3ng/ml, respectively. The maximum concentration (C_{max}), area under the curve (AUC) and amount excreted in the urine for both Naltrexone and 6- β -naltrexol increased proportionally as the amount of Naltrexone administered increased from 50-200mg. The time to maximum concentration (T_{max}) is 1 hour for both Naltrexone and 6- β -naltrexol. The elimination half-life ($T_{1/2}$) values for Naltrexone and 6- β -naltrexol 3.9 and 12.9 hours, respectively. The mean elimination half-life ($T_{1/2}$) and time to maximum concentration (T_{max}) for naltrexone and 6- β -naltrexol are independent of dose.

Naltrexone does not accumulate during chronic dosing. As predicted by its longer half-life, plasma levels of 6- β -naltrexol increase by 40% during chronic Naltrexone dosing. The total body clearance of Naltrexone is 1.5 L/min, which approximates liver blood flow and suggests naltrexone is a highly extracted compound. A renal clearance of 127 mL/min for naltrexone suggests it is solely cleared by glomerular filtration. A renal clearance of 283 mL/min for 6- β -naltrexol suggests an additional renal tubular secretory mechanism. The volume of distribution for naltrexone following IV administration is estimated to be 1350 L. In vitro tests with human plasma show Naltrexone to be 21% bound to plasma protein over the therapeutic dose range.

INDICATION: To provide blockade of the pharmacologic effects of exogenously administered opioids as an adjunct to the maintenance of the opioid-free state in detoxified, formerly opioid-dependent individuals.

RECOMMENDED DOSAGE:

Induction of Naltrexone therapy: Do not attempt treatment until Naloxone challenge is negative. Initiate treatment with Naltrexone using the following guidelines:

1. Treatment should not be attempted until the patient has remained opioid-free for 7-10 days. Self-reporting of abstinence from opioids should be verified by analysis of the patient's urine for absence of opioids. The patient should not be manifesting withdrawal signs or reporting withdrawal systems.
2. A Naloxone challenge test should be administered to patient. If signs of opioid withdrawal are still observed following Naloxone challenge, treatment with Naltrexone should not be attempted. The Naloxone challenge can be repeated in 24 hours
3. Treatment should be initiated carefully slowly increasing the dose of naltrexone administered. This can be accomplished by administration of 25mg of Naltrexone initially. The patient should be observed for 1 hour. If no withdrawal signs occur, the patient may be given the rest of the daily dose.

Naloxone Challenge Test: The Naloxone challenge test should not be performed in a patient showing clinical signs or symptoms of opioid withdrawal, or in a patient whose urine contains opioids.

The Naloxone challenge test may be administered by either the IV or SC routes. IV Challenge: Following observation of the patient, 0.8mg of Naloxone should be drawn into a sterile syringe. If the IV route of administration is selected, 0.2mg of Naloxone should be injected, and while the needle is still in the patient's vein, the patient should be observed for 30 sec for evidence of withdrawal signs or symptoms. If there is no evidence of withdrawal, the remaining 0.6mg of Naloxone should be injected and the patient observed for an additional period of 20 min for signs and symptoms of withdrawal.

SC Challenge: If the SC route is selected, 0.8mg should be administered SC and the patient observed for signs and symptoms of withdrawal for 45 min.

Conditions and Technique for Observation of Patient: During the appropriate period of observation, the patient's vital signs and signs of withdrawal symptoms should be monitored. It is also important to question the patient carefully. The signs and symptoms of opioid withdrawal include, but are not limited to the following:

Withdrawal Signs: Stiffness or running nose, tearing, yawning, sweating, tremor vomiting or piloerection. Withdrawal Symptoms: Feeling of temperature change, joint or bone and muscle pain, abdominal cramps, skin crawling etc.

Interpretation of the challenge: Warning: The elicitation of the enumerated signs or symptoms indicate a potential risk for the subject and Naltrexone should not be administered. If no signs or symptoms of withdrawal are observed, Naltrexone may be administered. If there is any doubt in the observer's mind that the patient is not in an opioid-free state or is in continuing withdrawal, Naloxone should be re-administered as follows:

Confirmatory re-challenge (if necessary): 1.6mg of Naloxone should be injected IV and the patient again observed for signs and symptoms of withdrawal. If none are present, Naltrexone may be administered. If signs and symptoms of withdrawal are present, administration of Naltrexone should be delayed until repeated Naloxone challenge indicates the patient is no longer at risk.

Maintenance Treatment: Once the patient has been started on Naltrexone, 50mg every 24 hours will produce adequate clinical blockade of the actions of parenterally administered opioids (i.e. this dose will block the effects of a 25mg IV heroin challenge). A flexible approach to a dosing regimen may be employed. Thus patients may receive 50mg of Naltrexone every weekday with a 100mg dose on Saturday, or patients may receive 100mg every other day, or 150mg every 3rd day. While the degree of opioid blockade may be reduced by using higher doses at longer dosing intervals, improved patient compliance may result from dosing every 48-72 hours.

Many have employed the following dosing regimen: 100mg on Monday 100mg on Wednesday and 150mg on Friday. This dosing schedule appeared to be acceptable to many Naltrexone patients successfully maintaining their opioid-free state.

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS: Patients receiving opioid analgesics, opioid-dependent patients and patients in acute opioid withdrawal (see Warnings). Any individual who has failed to pass the Naloxone challenge (see Dosage & Administration). Any individual who has a positive urine screen for opioids. Any individual with a history of sensitivity to naltrexone. It is not known if there is any cross-sensitivity with naloxone or other phenanthrene containing opioids. Any individual with acute hepatitis or liver failure.

WARNINGS & PRECAUTIONS:

Hepatotoxicity

Naltrexone has the capacity to cause dose – related hepatocellular injury. Therefore the physician should establish whether the patient has subclinical liver injury or disease before initiate treatment with Naltrexone. (See Laboratory Tests under Precautions). Naltrexone is contraindicated in acute hepatitis or liver failure, but its use even in patients with evidence of less severe liver disease or a history of recent liver disease must be carefully considered in light of its hepatotoxin potential. The evidence that identified Naltrexone as a hepatotoxin was not obtained in its use at the doses recommended for opiate blockade, where the changes in serum levels of liver enzymes seen were similar to those present at baseline. However, the margin of separation between the apparently safe and the hepatotoxic doses appears to be only 5-fold or less.

Evidence of Naltrexone's hepatotoxic potential is derived when Naltrexone was administered to obese subjects at a dose approximately 5-fold that recommended for the blockade of opiate receptors (300mg/day). 5 of 26 naltrexone recipients developed elevations of serum transaminases (i.e. peak SGPT values ranging from a low of 121 to a high of 523; 3-19 times their baseline values) after 3-8 weeks treatment. Although the patients involved were generally clinically asymptomatic and the transaminase levels of all patient on whom follow-up was obtained returned to (or toward) baseline values in a matter of weeks, the lack of any transaminase elevations of similar magnitude in any of the 24 placebo patients study is persuasive evidence the naltrexone is a direct (i.e. Not an idiosyncratic) hepatotoxin. This conclusion is also supported by evidence from other placebo-controlled exposures to Naltrexone at doses from 1- to 2- fold the amount recommended for opiate blockade consistently produce more numerous and more significant elevations of serum transaminases that did placebo, and reports of transaminase elevations in 3 to 9 patients with Alzheimer's Disease who received Naltrexone (u-p to 300mg/day) for 5-8 weeks.

Unintended Precipitation of Abstinence: To prevent occurrence of an acute abstinence syndrome, or exacerbation of a preexisting subclinical abstinence syndrome, patients should remain opioid-free for a minimum of 7-10 days before starting naltrexone. Since the absence of a opioid drug in the urine is often not sufficient proof that a patients is opioid-free, a Naloxone challenge should be employed to exclude the possibility of precipitating a withdrawal reaction following administration of Naltrexone. The Naloxone challenge test is described in Dosage & Administration. While Naltrexone is a potent antagonist with a prolonged pharmacologic effect (24-72 hours), the blockade produced by Naltrexone is surmountable. This is useful in patients who may require analgesia, but possess a potential risk to individuals who attempt, on their own, to overcome the blockade by administering large amounts of exogenous opioids. Indeed any attempt by a patient to overcome the antagonism by taking opioids is very dangerous and may lead to a fatal overdose. Injury may arise because the plasma concentration of exogenous opioids attained immediately following their acute administration may be sufficient to

overcome the competitive blockade. As a consequence, the patient may be in immediate danger of suffering life-endangering opioid intoxication (e.g. respiratory arrest, circulatory collapse). Also lesser amounts of exogenous opioids may prove dangerous if they are taken in a manner (i.e. relatively long after last dose of naltrexone) and in an amount so that they persist in the body longer than effective concentrations of naltrexone and its metabolites. Patients should be told of the serious consequences of surmounting the opiate blockade: See Information for Patients under Precautions.

General: Actions suggested when reversal of Naltrexone blockade is required: In an emergency situation requiring analgesia which can only be achieved with opioids, the amount of opioid required may be greater than usual, and the resulting respiratory depression may be deeper and more prolonged. No methods to reverse overdose have been established. Therefore in such circumstances, a rapidly acting analgesic which minimizes respiratory depression is preferred. The amount of analgesic administered should be titrated to the needs of the patient. Additionally, non-receptor mediated actions may occur (e.g. facial swelling, itching, generalized erythema presumably due to histamine release). Irrespective of the drug chosen to reverse Naltrexone (naltrexone HCl) blockade, the patient should be monitored closely by appropriately trained personnel in a hospital setting.

Actions suggested when withdrawal is accidentally precipitated with Naltrexone: Severe opioid withdrawal syndromes precipitated by the accidental ingestion of Naltrexone have been reported in opioid-dependent individuals. Symptoms of withdrawal have usually appeared within 5 min of ingestion of Naltrexone and have lasted for up to 48 hours. Mental status changes including confusion, somnolence and visual hallucinations have occurred. Significant fluid losses from vomiting and diarrhea have required IV fluid administration. In all cases, patients were closely monitored and therapy tailored to meet individual requirements.

Information for Patients: It is suggested that the prescribing physician relate the following information to patients being treated with Naltrexone. Naltrexone has been prescribed as part of the comprehensive treatment for drug dependence. The patient should carry identification card to alert medical personnel to the fact that Naltrexone is taken. A Naltrexone medication card may be obtained from physician and can be used for this purpose. Carrying the identification card should help to ensure that the patient can obtain adequate treatment in an emergency. If medical treatment is required, the treating physician should be told of Naltrexone therapy being received. Naltrexone should be taken as directed by the physician. Any attempt to self-administer heroin or any other opiate drug in small doses, will not perceive any effect. Most important, an attempt to self-administer large doses of heroin or any other narcotic may lead to death or sustain serious injury including coma.

Laboratory Tests: Test designed to detect hepatic injury should be obtained prior to initiation of Naltrexone therapy and periodically thereafter (see Hepatotoxicity under Warnings).

Periodic testing of all patients after initiation of treatment is critical if the occurrence of Naltrexone-induced liver damage is to be detected at the earliest possible time. Evaluations of tests to detect liver injury are recommended on a monthly basis during the first 6 months of use; thereafter, clinical judgment about the frequency of monitoring must be relied upon. Laboratory tests which may be used for the separation and detection of morphine, methadone or quinine in the urine and with which Naltrexone does not interfere include thin layer, gas liquid and high pressure liquid chromatographic methods.

Carcinogenicity: Carcinogenesis: In a 2 year carcinogenicity in rats. There were small increases in the numbers of mesotheliomas in males and tumors of vascular origin in both sexes. The numbers of tumors were within the range seen in historical control groups, except for the vascular tumors in females, where the 4% incidence exceeded the historical maximum of 2%.

Mutagenicity: Of 22 distinct test using bacterial, mammalian and tissue culture systems, all tests were negative except for weakly positive findings in the Drosophila recessive lethal assay and nonspecific DNA repair tests with *E. coli*. The significance of these findings is undetermined.

Impairment of Fertility: Naltrexone (100mg/kg, approximately 140 times the human therapeutic dose) caused a significant increase in pseudopregnancy in the rate. A decrease in the pregnancy rate of mated female rats also occurred. The relevance of these observations to human fertility is not known.

Use in children: The safe use of Naltrexone in subjects younger than 18 years has not been established.

INTERACTION WITH OTHER MEDICAMENTS:

Interference with the action of narcotic-containing drug product: Patients taking Naltrexone may not benefit from opioid-containing medicines, e.g. cough and cold preparations antidiarrheal preparations and opioid analgesics. Where a non-opioid-containing alternative is available, it should be used.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Naltrexone has been shown to have an embryocidal effect in the rat and rabbit when given in doses approximately 140 times human therapeutic dose. This effect was demonstrated in rats dosed with Naltrexone (100mg/kg) prior to and throughout gestation, and rabbits treated with 60mg/kg of Naltrexone during the period of organogenesis. There are no adequate and well-controlled studies in pregnant women. Naltrexone should be used in pregnancy only when the potential benefit justifies the potential risk to the fetus.

Labour and Delivery: Whether or not Naltrexone affects the duration of labour and delivery is known.

Use in Lactation: Whether or not Naltrexone is excreted in human milk is unknown. Caution should be exercised when Naltrexone is administered to a nursing mother.

SIDE EFFECTS: Naltrexone in detoxified, formerly opioid-dependent individuals failed to identify any single, serious untoward risk of Naltrexone use, placebo-controlled employing up to 5-fold higher doses of Naltrexone (up to 300 mg/day) than that recommended for use in opiate receptor blockade have shown that Naltrexone cause hepatocellular injury in a substantial proportion of patients exposed at this higher dose (see Laboratory Test under Precautions). Available evidence does not incriminate Naltrexone, used at any dose, as a cause of any other serious untoward event for patients who is "opioid-free". It is critical to recognize that naltrexone can precipitate or exacerbate abstinence signs and symptoms in any individual who is not completely free of exogenous opioids.

Naltrexone used at the doses recommended to produce opiate receptor blockade does not appear to be the cause of any numerous adverse events including liver function abnormalities, in individuals with a history of both alcohol and substance abuse.

Untoward events observed (e.g. lymphocytosis, transaminase elevation, GI disturbances) were findings that would be anticipated in any similar population not treated with Naltrexone. Baseline abnormalities, including transaminase elevations, improved or returned to normal during the course of treatment with naltrexone. In summary, among opioid-free individuals, naltrexone administration at the recommended dose has not been associated with a predictable profile of serious adverse or untoward events. However, as previously-mentioned among individuals using opioids, Naltrexone may cause serious reactions (see contraindications, warnings, Dosage & Administration).

The following adverse reactions have been reported both at baseline and during the Naltrexone medication period at an:

Incidence rate of >10%: Difficulty in sleeping, anxiety, nervousness, abdominal pain, cramps, nausea and/or vomiting, low energy, joint and muscle pain and headache.

The incidence was <10% for: Loss of appetite, diarrhea, constipation, increase thirst, increased energy, feeling down, irritability, dizziness, skin rash, delayed ejaculation, decreased potency and chills.

The following events occurred in <1% of subjects:

Respiratory: Nasal congestion, itching, rhinorrhea, sneezing, sore throat, excess mucus or phlegm, sinus trouble, heavy breathing, hoarseness, cough, shortness of breath.

Cardiovascular: Nose bleeds, phlebitis, edema, increased blood pressure, non-specific ECG changes, palpitations, tachycardia.

Gastrointestinal: Excessive gas, hemorrhoids, diarrhea, ulcer.

Musculoskeletal: Painful shoulders, legs or knees: tremor, twitching.

Genitourinary: Increased frequency of or discomfort during urination; increased or decreased sexual interest.

Dermatologic: Oily skin, pruritus, acne, athlete's foot, cold sores, alopecia.

Psychiatric: Depression, paranoia, fatigue, restlessness, confusion, disorientation, hallucinations, nightmares, bad dreams.

Special Senses: Eyes (blurred, burning, light sensitive, swollen, aching, strained). Ears ("clogged" aching, tinnitus)

General: Increased appetite, weight loss weight gain, yawning, somnolence, fever, dry mouth, head "pounding", inguinal pain, swollen glands, "side" pains, cold feet, "hot spells" Lethargy and somnolence have been reported following dosing of Naltrexone and thioridazine.

Laboratory Tests: With the exception of liver test abnormalities (see Warning, Precautions, etc), adverse reactions reports have not shown consistent patterns of abnormalities that can be attributed to treatment with Naltrexone.

Naltrexone for the blockade of opiate receptors, abnormal liver function tests and lymphocytosis were the 2 most common categories of abnormalities reported. As noted earlier, these abnormalities are common among populations of parenteral opioid users and alcoholics.

Idiopathic thrombocytopenic purpura was reported in one patient who may have been sensitized to Naltrexone in a previous course of treatment with Naltrexone. The condition cleared without sequelae after discontinuation of naltrexone and corticosteroid treatment.

Drug Abuse and Dependence: Naltrexone is a pure opioid antagonist. It does not lead to physical or psychological dependence. Tolerance to the opioid antagonist effect is not known to occur.

SYMPTOMS AND TREATMENT OF OVERDOSE: There is no clinical experience with Naltrexone overdose in humans.

Treatment: Consideration should be given to contacting a poison control center for the most up-to-date information. In view of the lack of actual experience in the treatment of Naltrexone overdose, patients should be treated symptomatically in closely supervise environment.

STORAGE CONDITIONS:

Store below 30°C. Protect from light.

Keep away from children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Blister pack of 10 x 10's.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.